Course Notes

Combinatorics II

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目录

第一章 Lecture 1 **2**
1.1 Introduction to Graphs 2

第二章 Lecture 2 **16**
2.1 Tree 21

第三章 Lecture 3 **27**
3.1 Chrometic number 27
3.2 Ramsey theorem 30

第四章 Homework **36**
4.1 Week 1 36

第一章　Lecture 1

Basic Information

Office hours: Mon, 13:45-15:45.

Main book: *A course in combinatorics* by J.H. van Lint and R.M. Wilson, 2nd edition. And the lecture notes by Prof. Ihringer.

Grade: Final Exam: 40%, midterm 30 % (April 15, week 8 in class), Homework 20%, Quizzes 10% (very easy based on HW).

Homework:

- Biweekly
- Only handwritten on paper
- Submission in class
- Only in English

Preliminary Schedule:

- **Week 1-4:** Basic graph theory, including trees, Ramsey numbers, Turan numbers, Hall's marriage theorem, etc.
- **Week 5-6:** Hadamard matrices.
- **Week 6-8:** Projective spaces.
- **Week 9:** Designs.
- **Week 10:** Strongly regular graphs.
- **Week 11:** Generalized Quadrangles.
- **Week 12-13:** General Spectral Graph Theory.
- **Week 14:** Associated Schemes.
- **Week 15:** Coding Theory.

1.1 Introduction to Graphs

Definition 1.1.1. *A graph G is a pair (V, E) where V is a set of vertices and E is a set of edges.*

Let $G = (V, E)$ be a graph with vertex set V and edge set E.

- If G is directed, then edges are ordered pairs and $E \subseteq V \times V$;
- If G is simple undirected, then edges are unordered pairs and $E \subseteq \binom{V}{2}$. (We call such G a normal graph in our course.)

Definition 1.1.2. *Let $G = (V, E)$ be a graph.*

- *Let $e = \{x, y\} \in E$. The vertices x and y are said to be* ***incident*** *with the edge e. Two vertices x, y are called* ***adjacent*** *if $\{x, y\} \in E$.*
- *A vertex is called an* ***isolated vertex*** *if it is not incident with any edge.*
- *An edge connecting a vertex to itself is called a* ***loop****.*
- *Two or more edges connecting the same pair of vertices are called* ***multiple edges****.*

Example 1.1.3. *The following figure illustrates the definitions:*

Incident / Adjacent *Isolated Vertex* *Loop* *Multiple Edges*

- ***Incident/Adjacent****: Vertices x and y are adjacent and incident to edge e.*
- ***Isolated****: Vertex z has no incident edges.*
- ***Loop****: Vertex u has an edge connecting to itself.*
- ***Multiple Edges****: Vertices v and w are connected by two edges.*

Definition 1.1.4. *A graph is called a* ***planar graph*** *if it possesses a planar drawing, that is, it can be drawn in the plane such that no two edges intersect except at their endpoints.*

Example 1.1.5. *Consider the following two drawings:*

G_1 G_2

G_1 and G_2 are ***isomorphic****: If we pull the center node of G_1 then we get G_2. G_1 is planar, but G_2 is not. We call G_1 the plannar drawing of G_2, which means that G_1 has no edges intersect.*

Definition 1.1.6. *A graph without loops and multiple edges is called a* ***simple graph****.*

Remark 1.1.7. *In the two statements above, the intended notion is essentially the same. The condition "G is simple undirected" together with $E \subseteq \binom{V}{2}$ means that every edge is an unordered pair $\{u,v\}$ of* distinct *vertices, so loops are excluded automatically, and since E is a set (rather than a multiset), multiple edges are excluded as well. Hence this coincides with the standard definition: a* simple graph *is a graph with no loops and no multiple edges.*

In our course notes, such graphs are called normal graphs*. The only minor caveat is that some texts use "simple graph" with the convention that graphs are undirected by default, whereas others may state "simple undirected" explicitly. In the present context, these conventions agree.*

Convention: All our graphs are simple. Unless we say otherwise.

Definition 1.1.8 (Map coloring and the associated graph)**.** *A* map *(on the plane or on the sphere) is a decomposition of the surface into finitely many regions ("countries") such that any two regions meet either in a boundary arc, in finitely many boundary points, or not at all. Two countries are said to be* adjacent *if they share a boundary segment of positive length (meeting at a single point does* not *count). Given such a map, we define its* adjacency graph *(also called the* dual graph*) $G = (V,E)$ as follows:*

$$V = \{\textit{countries of the map}\}, \qquad \{u,v\} \in E \iff \textit{countries } u \textit{ and } v \textit{ are adjacent.}$$

Remark 1.1.9 (Planarity)**.** *The adjacency graph G is* planar*: it can be drawn in the plane with no edge crossings by placing a vertex inside each country and drawing an edge between two vertices through the common boundary segment of the corresponding countries. Thus the problem of coloring maps is equivalent to coloring planar graphs.*

Definition 1.1.10 (Vertex coloring and chromatic number)**.** *Let $G = (V,E)$ be a graph. A* (proper) vertex coloring *of G with k colors is a map $c : V \to \{1,2,\dots,k\}$ such that*

$$\{u,v\} \in E \implies c(u) \neq c(v).$$

The smallest k for which such a coloring exists is called the chromatic number *of G and is denoted by $\chi(G)$.*

Theorem 1.1.11 (Four-Color Theorem)**.** *For every planar graph G, one has*

$$\chi(G) \leq 4.$$

Equivalently, every planar map can be colored with at most four colors so that any two adjacent countries receive different colors.

Remark 1.1.12 (Why "at most 5 colors" is easy)**.** *A classical result (the* Five-Color Theorem*) states that every planar graph satisfies $\chi(G) \le 5$. This theorem admits a relatively short proof using standard planar graph tools such as Euler's formula and a reducible configuration argument (often presented via induction on $|V|$). By contrast, the Four-Color Theorem is substantially deeper and historically required extensive case-checking (computer-assisted) in its first proofs.*

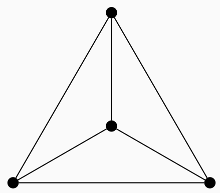
The following statements are equivalent:

$$\{x, y\} \in E \Leftrightarrow x \sim y \Leftrightarrow x, y \text{ adjacent } \Leftrightarrow x, y \text{ are neighbors}$$

Example 1.1.13. $|V| = n$, $E = \binom{V}{2}$. *Complete graph K_n.*

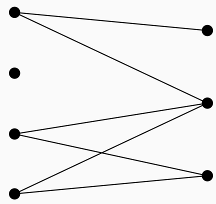
K_1 K_2 K_3 K_4 K_5

Remark: K_4 is a planar graph, but K_5 is not.

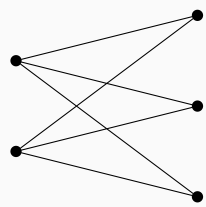


Planar drawing of K_4

Example 1.1.14. $G = (V_1 \cup V_2, E)$ *is bipartite if* $E \cap (\binom{V_1}{2} \cup \binom{V_2}{2}) = \emptyset$.

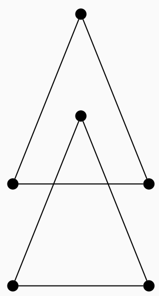


Example 1.1.15. *If $E = \{\{x, y\} : x \in V_1, y \in V_2\}$, then G is a complete bipartite graph, denoted by $K_{m,n}$ where $m = |V_1|, n = |V_2|$.*

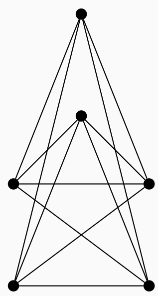


$K_{2,3}$

Example 1.1.16. *Similarly we can define k-partite graphs and complete k-partite graphs.*



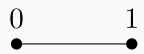
3-partite (2,2,2)
(not complete)



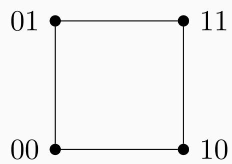
$K_{2,2,2}$

Example 1.1.17. *The* n***-dimensional hypercube*** Q_n *is the graph with vertex set* $V(Q_n) = \{0,1\}^n$*, the set of binary strings of length* n*. Two vertices* $x, y \in V(Q_n)$ *are adjacent if and only if they differ in exactly one coordinate (i.e., their Hamming distance is* 1*). The graph* Q_n *is* n*-regular with* $|V(Q_n)| = 2^n$ *and* $|E(Q_n)| = n2^{n-1}$*. For* $n = 3$*, we have* $110 \sim 111$*, whereas* $111 \nsim 100$*.*

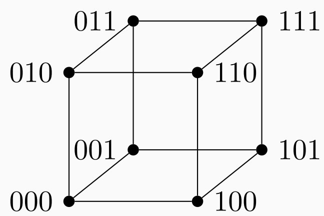
Q_0



Q_1

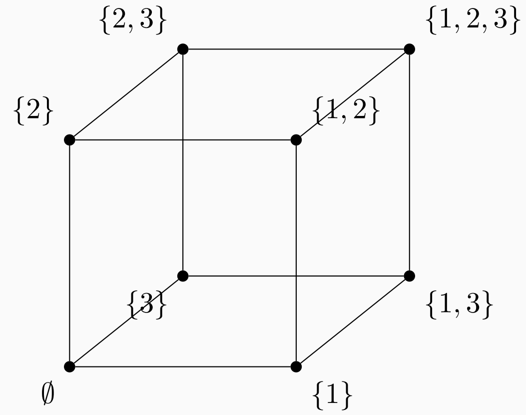


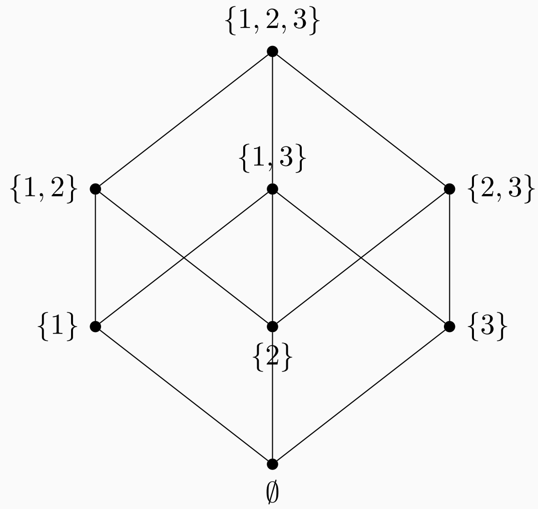
Q_2



Q_3

Example 1.1.18. *Let* S *be a set with 3 elements, for instance* $S = \{1,2,3\}$*. The power set* $\mathcal{P}(S)$ *can be made into a graph by defining two subsets* $A, B \subseteq S$ *to be adjacent if their symmetric difference has size one, i.e.,* $|A \Delta B| = 1$*. This graph is isomorphic to the 3-dimensional hypercube* Q_3*. The structure is often drawn as the Hasse diagram of the subset lattice.*

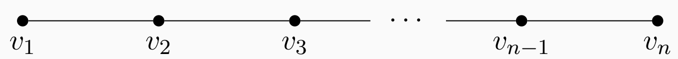




Hasse lattice of $\mathcal{P}(\{1,2,3\})$

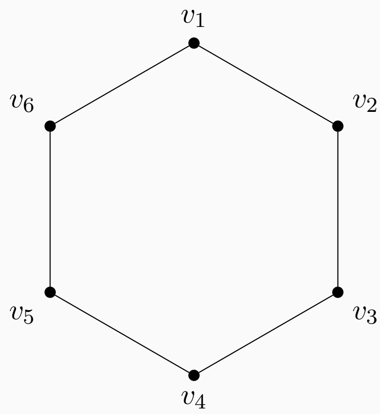
This graph is another representation of Q_3*.*

Example 1.1.19. P_n*: path of length* $n-1$*.*



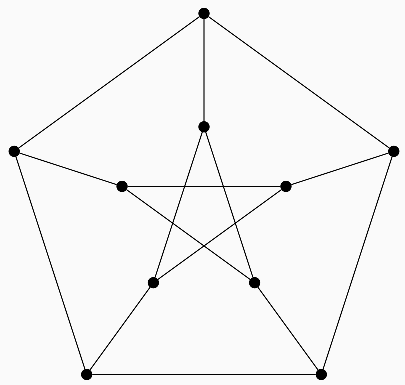
$$|E(P_n)| = n-1$$

Example 1.1.20. *With* $\{v_n, v_1\}$ *added, we get a cycle* C_n*. For example* C_6*:*



$|V(C_6)| = |E(C_6)| = 6$

Example 1.1.21. *Peterson Graph:*



Petersen graph

Each vertix has degree 3 so this graph is 3*-regular. Any two vertices linked by an edge has no common neighbor. Any 2 vertices not linked by an edge has a common neighbor.*
It is a strongly regular graph with parameters $(10, 3, 0, 1)$*.*

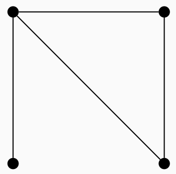
Definition 1.1.22 (Strongly regular graph)**.** *Let* $G = (V, E)$ *be a simple undirected graph. We say that* G *is* strongly regular *with parameters* (v, k, λ, μ) *if the following hold:*

1. $|V| = v$ *and* G *is* k*-regular (i.e. every vertex has degree* k*);*

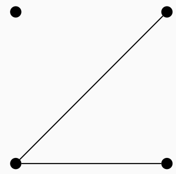
2. *for any two* adjacent *vertices* $x \sim y$*, the number of common neighbors of* x *and* y *is exactly* λ*, i.e.* $|\Gamma(x) \cap \Gamma(y)| = \lambda$*;*

3. *for any two* non-adjacent *vertices* $x \not\sim y$*, the number of common neighbors of* x *and* y *is exactly* μ*, i.e.* $|\Gamma(x) \cap \Gamma(y)| = \mu$*.*

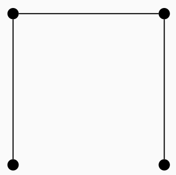
Example 1.1.23. *$G=(V,E)$, $\bar{G}=(V,\bar{E})$, where $\bar{E}=\binom{V}{2}\setminus E$. $\bar{G}$ is the complement of G.*



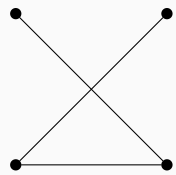
G



$\overline{G}$



P_4



$\overline{P_4}$

In this example, observe that $P_4 \cong \overline{P_4}$. We call such a graph self-complementary. A graph G is self-complementary if $G \cong \overline{G}$.

Definition 1.1.24. *Let $G=(V,E)$ and $G'=(V',E')$. They are isomorphic if and only if there exists a bijection $\phi: V \to V'$ such that $\{x,y\}\in E \Leftrightarrow \{\phi(x),\phi(y)\}\in E'$.*

Definition 1.1.25. *$\sharp y : y \sim x$ = degree of x, denoted by $\deg(x)$ or $\mathrm{d}(x)$. If $\mathrm{d}(x)=\mathrm{d}(y)$ for all $x,y\in V$, then G is d-regular.*

Remark 1.1.26 (Degree sequence: definition and uses)**.** *Let $G=(V,E)$ be a simple undirected graph with $n=|V|$. The* degree sequence *of G is the list of vertex degrees arranged in nonincreasing order,*

$$d_1 \geq d_2 \geq \cdots \geq d_n, \qquad \text{where } \{d_1,\ldots,d_n\}=\{\deg(v): v\in V\}.$$

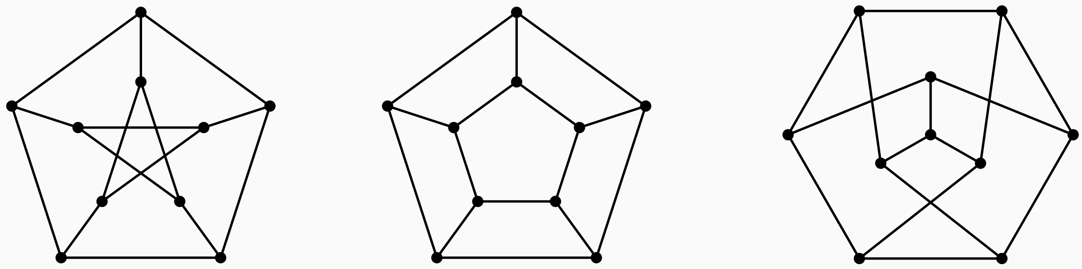
It is useful for:

- ***Coarse structural information:** by the Handshaking Lemma, $\sum_{i=1}^{n} d_i = 2|E|$, so the degree sequence determines the number of edges and reveals properties such as regularity (all d_i equal).*
- ***Realizability tests:** deciding whether a given integer sequence is* graphical *(i.e. occurs as the degree sequence of a simple graph), using criteria such as the Erdős–Gallai theorem or the Havel–Hakimi algorithm (which can also construct a realizing graph).*

- ***Isomorphism invariant:*** *isomorphic graphs have the same degree sequence, so differing degree sequences imply non-isomorphism, though the converse need not hold.*
- ***Algorithmic and modeling input:*** *degree sequences serve as compact summaries in extremal graph theory, network models (e.g. random graphs with prescribed degrees), and matrix-based methods (e.g. Laplacian $L = D - A$ where D is the degree matrix).*

Example 1.1.27. $d_1 = 2, d_2 = 1, d_3 = 1$. *This graph is* P_3.

Example 1.1.28. *Consider the following 3 graphs:*



G_1　　G_2　　G_3

$G_1 \ncong G_2$, $G_2 \ncong G_3$, *but* $G_1 \cong G_3$.

The *graph isomorphism problem* (GI) asks: given two finite graphs G and H, decide whether there exists a bijection between their vertex sets that preserves adjacency (i.e. an isomorphism). From the viewpoint of worst-case complexity, GI has long occupied an intermediate position: it is in NP, but it is neither known to be NP-complete nor known to admit a polynomial-time algorithm. A major milestone is Babai's 2015 breakthrough, which gives a *quasi-polynomial* time algorithm for GI, running in time $\exp\big((\log n)^{O(1)}\big)$ on n-vertex graphs, substantially improving the best previously known general upper bounds.

In parallel with the worst-case theory, there is a large body of *practical* work aimed at solving typical instances efficiently, often via *canonical labeling* and computation of automorphism groups. Since the 1990s, McKay's software `nauty` (and later `Traces`, with further algorithmic refinements) has been highly effective on a wide range of graphs encountered in applications. However, strong empirical performance does not settle the worst-case complexity question, and there exist carefully constructed families of graphs that can be difficult for certain refinement-and-search heuristics. Thus, it is important to distinguish clearly between (i) the unresolved *theoretical* question of whether GI is solvable in polynomial time in the worst case, and (ii) the largely successful but still evolving *algorithm engineering* of fast GI solvers for real-world inputs.

Proposition 1.1.29 (handshaking lemma). *$G = (V, E)$. Then*

$$\sum_{v \in V} \deg(v) = 2|E|$$

证明. Count $(x, e) \in V \times E$ where $x \in e$ in two ways. First, for each $e \in E$, there are exactly two vertices $x \in e$. So the number of such pairs is $2|E|$. Second, for each $x \in V$, there are exactly $\deg(x)$ edges $e \in E$ containing x. So the number of such pairs is $\sum_{v \in V} \deg(v)$. Equating these two counts gives the desired result. □

Theorem 1.1.30. *A finite graph (we only talk about finite graphs in our class) has an even number of vertices with odd degree.*

证明. Consider

$$\sum_{v \in V} \deg(v) = 2|E|,$$

which is an even integer.

Reduce this congruence modulo 2. Vertices of even degree contribute 0 modulo 2, and vertices of odd degree contribute 1 modulo 2. Hence

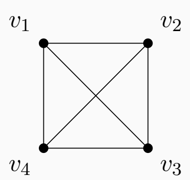
$$\sum_{v \in V} \deg(v) \equiv \#\{v \in V : \deg(v) \text{ is odd}\} \pmod 2.$$

Since the left-hand side is even, the right-hand side is $0 \pmod 2$, so the number of odd-degree vertices is even. □

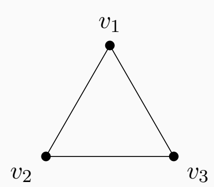
Definition 1.1.31 (subgraph). *Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is a* subgraph *of G if $V' \subseteq V$ and $E' \subseteq E$.*

Definition 1.1.32 (induced subgraph). *Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is an* induced subgraph *of G if $V' \subseteq V$ and $E' = E \cap \binom{V'}{2}$.*

Example 1.1.33. *Consider the graph K_4, we have two subgraph C_3 and C_4. The subgraph C_3 is an induced subgraph, but the subgraph C_4 is not.*

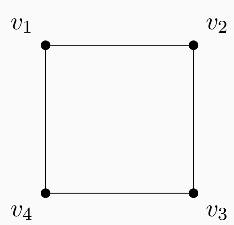


K_4



C_3

induced subgraph



C_4

not induced in K_4

For C_3, we choose the vertices $\{v_1, v_2, v_3\}$, and every edge among them in K_4 is kept, *so it is induced. For C_4, we use all four vertices but delete the two diagonals, so it is a subgraph but not an induced subgraph.*

Adjacency list representation.

Let $G = (V, E)$ be a (finite) graph with $V = \{1, 2, \dots, n\}$. In an *adjacency list* representation, the computer stores an array (or map) Adj indexed by vertices, where $\mathrm{Adj}[u]$ is the list of all neighbors of u:

$$\mathrm{Adj}[u] = \{\, v \in V : \{u, v\} \in E \,\} \quad \text{(undirected)}, \mathrm{Adj}[u] = \{\, v \in V : (u, v) \in E \,\} \quad \text{(directed)}.$$

For an undirected simple graph each edge $\{u, v\}$ appears twice, once in $\mathrm{Adj}[u]$ and once in $\mathrm{Adj}[v]$. This data structure supports efficient traversal of neighbors (e.g. BFS/DFS) in total time $O(|V| + |E|)$ and uses space $O(|V| + |E|)$ for sparse graphs.

For our human beings we use the following matrices:

Definition 1.1.34 (adjacent matrix)**.** *Let $G = (V, E)$ be a graph. Let $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, \dots, e_m\}$. Let $A = (a_{ij})$ where $a_{ij} = 1$ if $v_i \sim v_j$, and $a_{ij} = 0$ otherwise. Then A is called the* adjacent matrix *of G. For the base field usually we use $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}_2$, $\mathbb{F}_q$.*

Remark 1.1.35. *In our class we mainly use consider $A \in \mathbb{R}^{n \times n}$. Sometimes we will consider $A \in \mathbb{C}^{n \times n}$ when we move to spectral graph theory.*

Definition 1.1.36 (Incidence matrix)**.** *Let $B = (b_{ij})$, $B \in \mathbb{R}^{n \times m}$. $b_{ij} = 1$ if vertex v_i is incident to edge e_j, and $b_{ij} = 0$ otherwise. Then B is called the* incidence matrix *of G.*

Definition 1.1.37 (degree matrix)**.** *Let $D = (d_{ij})$ be a diagonal matrix where $d_{ii} = \deg(v_i)$, and $d_{ij} = 0$ if $i \neq j$. Then D is called the* degree matrix *of G.*

Exercise 1.1.38. *Prove that $A = BB^T - D$.*

证明. We assume throughout that $G = (V, E)$ is a finite *simple* undirected graph (no loops, no multiple edges) and that B is the (non-oriented) incidence matrix defined by

$$b_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is incident to } e_j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $M := BB^T = (m_{ik})$. Then

$$m_{ik} = (BB^T)_{ik} = \sum_{j=1}^{m} b_{ij}\, b_{kj}.$$

For fixed vertices v_i, v_k, the product $b_{ij} b_{kj}$ equals 1 precisely when *both* v_i and v_k are incident to the same edge e_j, and equals 0 otherwise. Hence m_{ik} counts the number of edges that are incident to both v_i and v_k.

Case 1: $i = k$. Then m_{ii} counts the number of edges incident to v_i, so

$$m_{ii} = \deg(v_i) = d_{ii}.$$

Case 2: $i \neq k$. In a simple graph, there is *at most one* edge joining v_i and v_k. Therefore

$$m_{ik} = \begin{cases} 1, & \text{if } v_i \sim v_k, \\ 0, & \text{otherwise,} \end{cases}$$

so for $i \neq k$ we have $m_{ik} = a_{ik}$.

Combining the two cases, we see that BB^T agrees with A on all off-diagonal entries, and has diagonal entries $\deg(v_i)$. Equivalently,

$$BB^T = A + D,$$

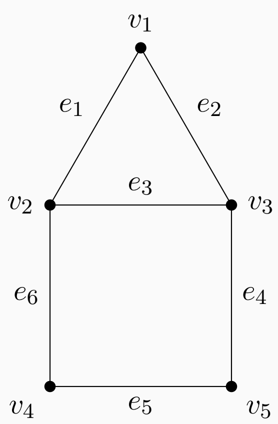
and therefore

$$A = BB^T - D,$$

as required. □

Definition 1.1.39 (Laplacian matrix)**.** *Let $L = D - A$, where D is the degree matrix and A is the adjacency matrix. Then L is called the* Laplacian matrix *of G.*

Example 1.1.40. *Consider this graph:*



With the vertex order $(v_1, v_2, v_3, v_4, v_5)$ and the edge order $(e_1, e_2, e_3, e_4, e_5, e_6)$, the *corresponding matrices are:*

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 \\ -1 & -1 & 3 & 0 & -1 \\ 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

We can also verify the exercise $A = BB^T - D$ directly:

$$BB^T = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix}.$$

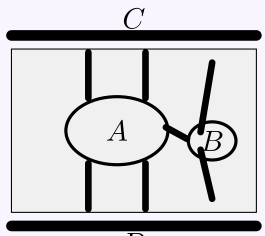
Hence

$$BB^T - D = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} = A.$$

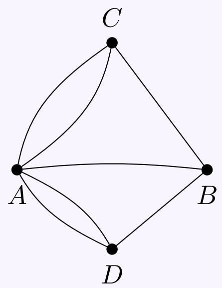
Definition 1.1.41 (k-walk)**.** *Let $G = (V, E)$ be a graph. A* k-walk *in G is a sequence of vertices $(v_0, v_1, \ldots, v_k)$ such that $\{v_i, v_{i+1}\} \in E$ for all $i = 0, 1, \ldots, k-1$. A walk* *is closed if $v_0 = v_k$. If a k-walk is a path, then all vertices are distinct. If a k-walk is a cycle, then all vertices are distinct except for the first and last vertices.*

第二章　Lecture 2

Topic 2.0.1 (The Seven Bridges of Königsberg)**.** *In Königsberg (now Kaliningrad), the river Pregel passed through the city with two islands and seven bridges connecting the islands and the two banks (see figure). The problem: can one start at some place, cross each bridge exactly once, and return to the start? Euler (1736) abstracted the two banks and two islands as four vertices and the seven bridges as seven edges, obtaining the graph G; he proved that such a closed trail exists if and only if every vertex has even degree. In G all four vertices have odd degree, so no solution exists. This problem is regarded as the origin of graph theory and Eulerian trails.*



Schematic of the seven bridges



Abstract graph G

Definition 2.0.2. *An Eulerian path is a walk which contains each edge of a graph exactly once.*

Definition 2.0.3. *An Eulerian cycle is a closed Eulerian path. A graph is called Eulerian if it contains an Eulerian cycle.*

Lemma 2.0.4. *Let $G = (V, E)$ be a graph with $|E| \geq 1$. If every vertex has degree at least 2 (i.e., $\deg(v) \geq 2$ for all $v \in V$), then G contains a cycle.*

证明. Take a maximal path $P = (v_0, v_1, \ldots, v_k)$ in G (so no edge extends P at either end). Since $|E| \geq 1$, we have $k \geq 1$. The vertex v_k has $\deg(v_k) \geq 2$, so v_k has a neighbor other than v_{k-1}. Any such neighbor must lie on P: if v_k were adjacent to some $u \notin \{v_0, \ldots, v_{k-1}\}$, then P could be extended to $v_0 \cdots v_k u$, contradicting maximality. So v_k is adjacent to v_i for some i with $0 \leq i \leq k-2$. Then $(v_i, v_{i+1}, \ldots, v_k, v_i)$ is a cycle in G. $\square$

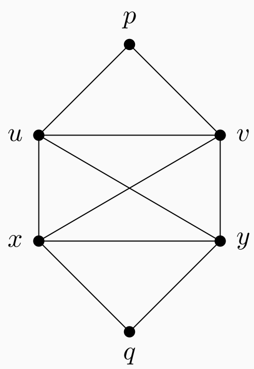
Theorem 2.0.5. *A finite connected graph is Eulerian if and only if every vertex has even degree.*

证明. *Necessity* ($\Rightarrow$). Let $C = (x_0, x_1, \ldots, x_n)$ be an Eulerian cycle with $x_0 = x_n$. Each edge is used exactly once. For any vertex $y \in V$, each time C passes through y it uses one edge to enter and one to leave; so the number of edges incident to y used by C is twice the number of times y appears in $(x_1, \ldots, x_{n-1})$. The start/end vertex $x_0 = x_n$ is left once and entered once, contributing 2 more. Thus $\deg(y)$ is even for all y.

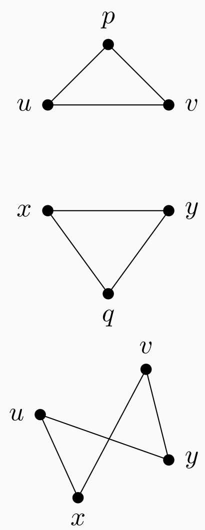
Sufficiency ($\Leftarrow$). (Hierholzer, 1873.) We use induction on $|E|$. If $|E| = 0$, then G is a single vertex and the trivial closed walk is an Eulerian cycle. Suppose $|E| \geq 1$. Since G is connected and every vertex has even degree, $\delta(G) \geq 2$, so G contains a cycle by the lemma above. Start at any vertex x and take a maximal walk that does not repeat any edge. Whenever the walk enters a vertex other than x, it has used an odd number of edges at that vertex so far, so an odd number (hence at least one) remains; thus the walk cannot get stuck except at x. So the walk returns to x, forming a closed trail C_1 with edge set $E_1 \subseteq E$.

Remove E_1 from G. In $G' = (V, E \setminus E_1)$, every vertex still has even degree (each vertex on C_1 lost 0 or 2 incident edges). So every non-trivial component of G' has an Eulerian cycle by the induction hypothesis. If $E_1 = E$, then C_1 is an Eulerian cycle. Otherwise, pick a vertex v that lies on C_1 and belongs to a component H of G' with at least one edge. That component has an Eulerian cycle C_2 (starting and ending at v). Splice C_2 into C_1 at v: traverse C_1 until v, then traverse all of C_2, then continue C_1 from v. Repeating this for every such component yields an Eulerian cycle of G. The algorithm runs in $O(|E|)$ time. □

Example 2.0.6. *Consider the graph below. On the left is the original connected graph; on the right we redraw the three edge-disjoint cycles that appear in the proof of Hierholzer's theorem.*



original graph



$C_1 = (u, x, v, y, u)$ *redrawn*

Choose the first closed trail to be

$$C_1 = (u, x, v, y, u),$$

which uses the four edges ux, xv, vy, and yu. This is the closed trail: start at u, keep following unused edges, and because every time we enter a vertex we can also leave it again, the trail cannot get stuck anywhere except at the starting vertex. Hence the walk comes back to u and closes up.

Now remove the edge set $E_1 = \{ux, xv, vy, yu\}$. The remaining graph $G' = (V, E \setminus E_1)$ has exactly two non-trivial components: the top triangle (u, p, v, u) and the bottom triangle (x, q, y, x). Every vertex in these components still has even degree, so by the induction hypothesis each component has its own Eulerian cycle.

Finally, splice those smaller cycles into C_1. Start to traverse C_1 at u, but before leaving u for the edge ux, first run through the top triangle (u, p, v, u). When the tour later reaches x, run through the bottom triangle (x, q, y, x), and then continue along the original trail. One Eulerian cycle of the whole graph is therefore

$$(u, p, v, u, x, q, y, x, v, y, u).$$

This illustrates the sentence "traverse C_1 until v, then traverse all of C_2, then continue C_1 from v": each smaller Eulerian cycle is inserted at a shared vertex, and after all insertions every edge of the original graph has been used exactly once.

Definition 2.0.7. *A Hamiltonion cycle is a closed walk which uses every vertex precisely once. A graph is called Hamiltonian if it contains a Hamiltonian cycle.*

Theorem 2.0.8. *Let $G = (V, E)$ be a simple graph on $n \geq 3$ vertices. If for every vertex $v \in V$, $\deg(v) \geq \frac{n+1}{2}$, then G is Hamiltonian.*

证明. First, we show that G is connected. Suppose for a contradiction that G is disconnected. Let C be the smallest connected component of G. The number of vertices in C is at most $\lfloor \frac{n}{2} \rfloor$. Thus, the maximum degree of any vertex in C is at most $\lfloor \frac{n}{2} \rfloor - 1 < \frac{n+1}{2}$, which contradicts the degree condition. Hence, G is connected.

We proceed by contradiction. Assume that G is not Hamiltonian. Let $P = v_1 v_2 \dots v_k$ be a longest path in G. Since G is connected and $n \geq 3$, we have $k \geq 3$.

Because P is a longest path, all neighbors of its endpoints v_1 and v_k must lie within the set $\{v_1, v_2, \dots, v_k\}$. If they had neighbors outside P, we could extend P to a longer path, contradicting its maximality.

Consider the sets of indices of the neighbors of v_1 and v_k on the path P:

$$S = \{i \in \{1, 2, \dots, k-1\} \mid \{v_1, v_{i+1}\} \in E\}$$
$$T = \{i \in \{1, 2, \dots, k-1\} \mid \{v_i, v_k\} \in E\}$$

Note that $|S| = \deg(v_1)$ and $|T| = \deg(v_k)$. By the given condition, we have:

$$|S| + |T| = \deg(v_1) + \deg(v_k) \geq \frac{n+1}{2} + \frac{n+1}{2} = n + 1$$

Since both S and T are subsets of $\{1, 2, \ldots, k-1\}$, and $k \leq n$, the maximum possible size of their union is $k - 1 \leq n - 1$.

By the Principle of Inclusion-Exclusion (or the Pigeonhole Principle), we have:

$$|S \cap T| = |S| + |T| - |S \cup T| \geq (n+1) - (n-1) = 2 > 0$$

Therefore, the intersection $S \cap T$ is non-empty. There exists some index i ($1 \leq i < k$) such that $\{v_1, v_{i+1}\} \in E$ and $\{v_i, v_k\} \in E$.

Using these two edges, we can construct a cycle C containing exactly the vertices of P:

$$C = v_1 v_2 \ldots v_i v_k v_{k-1} \ldots v_{i+1} v_1$$

Since we assumed G is not Hamiltonian, C does not contain all vertices of G, so $k < n$. Because G is connected, there must exist a vertex $u \notin V(C)$ that is adjacent to some vertex $v_j \in V(C)$.

Since v_j is on the cycle C, we can break the cycle by removing one of the edges incident to v_j in C, say $\{v_{j-1}, v_j\}$. We then add the edge $\{u, v_j\}$ to form a new path:

$$P' = u v_j v_{j+1} \ldots v_k \ldots v_1 \ldots v_{j-1}$$

The length of P' is $k + 1$, which strictly exceeds the length of P. This contradicts the assumption that P is a longest path in G.

Thus, our assumption that G is not Hamiltonian must be false. G contains a Hamiltonian cycle. □

Remark 2.0.9. *The statement "Random graphs are Hamiltonian" asserts that a random graph $G(n, p)$ contains a Hamiltonian cycle asymptotically almost surely (a.a.s.) provided the edge probability p exceeds a sharp threshold. Specifically, if*

$$p \geq \frac{\ln n + \ln \ln n + \omega(1)}{n},$$

then

$$\lim_{n \to \infty} \mathbb{P}(G(n,p) \textit{ is Hamiltonian}) = 1.$$

This contrasts with the NP-completeness of the Hamiltonian cycle problem for general graphs.

The Hamiltonian cycle problem is a fundamental example of an **NP-complete** problem. In the context of computational complexity:

- **P (Polynomial time)** represents the class of problems that can be solved efficiently (in polynomial time) by a deterministic computer (Turing machine).

- **NP (Nondeterministic Polynomial time)** consists of problems where a solution, once guessed, can be verified efficiently. Equivalently, these are solvable by a non-deterministic Turing machine in polynomial time.
- **NP-complete** problems are the "hardest" problems in NP. A problem is NP-complete if it is in NP and every other problem in NP can be transformed (reduced) into it in polynomial time.

If an efficient algorithm exists for any NP-complete problem, then efficient algorithms exist for all problems in NP (i.e., P = NP).

Graph theory is a fundamental branch of mathematics that studies the properties of graphs, which are mathematical structures used to model pairwise relations between objects. A graph consists of vertices (also called nodes or points) connected by edges (also called links or lines). The field has grown tremendously since Euler's solution to the Königsberg bridge problem in 1736, and now encompasses numerous subdisciplines:

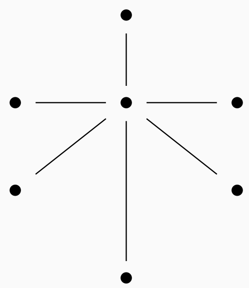
- **Structural graph theory** investigates the intrinsic properties and structural characterizations of graphs. Central topics include graph decompositions, minors, and the celebrated Graph Minor Theorem of Robertson and Seymour. This area explores fundamental questions about graph embeddings, planarity, and forbidden subgraph characterizations.
- **Algorithmic graph theory** focuses on the design and analysis of algorithms for graph problems. It addresses computational complexity, approximation algorithms, and efficient data structures for graph representation. Classic problems include shortest paths, maximum flow, graph coloring, and graph isomorphism. The interplay between structural properties and algorithmic efficiency is a key theme.
- **Extremal graph theory** studies the extremal (maximum or minimum) values of graph invariants under given constraints. A prototypical question is: what is the maximum number of edges in a graph on n vertices that does not contain a specified subgraph? The foundational Turán's theorem and the Erdős–Stone theorem are cornerstones of this field.
- **Spectral graph theory** explores connections between graph properties and the eigenvalues and eigenvectors of associated matrices (adjacency matrix, Laplacian matrix). It provides powerful algebraic tools for studying graph connectivity, expansion properties, and random walks on graphs.
- **Random graph theory** investigates properties of graphs generated by random processes, such as the Erdős–Rényi model $G(n, p)$. It studies phase transitions, threshold phenomena, and the typical behavior of graph parameters.
- **Algebraic graph theory** applies algebraic methods—including group theory, linear algebra, and representation theory—to study graphs. Topics include graph automorphisms, Cayley graphs, and association schemes.

These subfields are deeply interconnected and find applications across computer science, physics, biology, social network analysis, and numerous other disciplines.

2.1 Tree

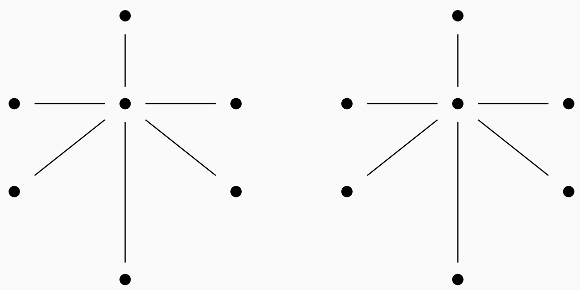
Definition 2.1.1. *A connected graph with no cycles is called a tree.*

Example 2.1.2. *This is an example of a tree. It is "木" in Chinese.*



Definition 2.1.3. *Disjoint unioin of trees is called a forest.*

Example 2.1.4. *This is an example of a forest. It is "林" in Chinese.*



Definition 2.1.5 (Spanning tree)**.** *Let $G = (V, E)$ be a connected (finite) graph. A* spanning tree *of G is a subgraph $T = (V, E_T)$ of G such that*

1. *$V(T) = V(G)$ (so T is spanning), and*

2. *T is a tree (equivalently: T is connected and contains no cycles).*

Equivalently, a spanning tree is a spanning, connected, acyclic subgraph of G (hence $|E_T| = |V| - 1$).

Proposition 2.1.6. *Every connected (finite) graph $G = (V, E)$ has a spanning tree.*

证明. Let $G = (V, E)$ be connected. Among all spanning subgraphs $H = (V, E_H)$ of G that are connected, choose one with $|E_H|$ minimal (such an H exists since E is finite and G itself is admissible). We claim that H is acyclic.

Indeed, if H contains a cycle C, pick an edge $e \in E(C)$ and let $H' := (V, E_H \setminus \{e\})$. Then H' is still connected, because the endpoints of e remain joined by the rest of the cycle, and any path using e can be rerouted along $C \setminus \{e\}$. This contradicts the minimality of $|E_H|$.

Hence H is connected and acyclic, so H is a tree. Since $V(H) = V(G)$, H is a spanning tree of G. □

Theorem 2.1.7. *The followings are equivalent:*

1. $G = (V, E)$ *is a tree.*
2. *Any two vertices are connected by a unique path.*
3. G *is connected with* $|V| - 1 = |E|$.

证明. (1) $\implies$ (2) If two vertices x, y are connected by two distinct paths, then their union contains a cycle.

(2) $\implies$ (1) If G contains a cycle, then any two distinct vertices on the cycle are connected by at least two different paths. Connectivity is given.

(1) $\implies$ (3) We proceed by induction on $|V|$. The base case $|V| = 1$ is trivial. For $|V| > 1$, let $P = (u, u_1, \ldots, u_l)$ be a longest path. Since G is a tree, all neighbors of the endpoint u must lie on P, which implies u is a leaf (degree 1). Removing this leaf and its incident edge leaves a tree on $|V| - 1$ vertices, and the result follows by induction.

(3) $\implies$ (1) Since G is connected, it has a spanning tree T. We know $|E(T)| = |V(T)| - 1 = |V(G)| - 1$. The given condition is $|E(G)| = |V(G)| - 1$. Since T is a subgraph of G, we must have $E(G) = E(T)$, which implies $G = T$. Thus, G is a tree.

□

Corollary 2.1.8. *Let G be a forest, then $\sharp$ connected components of $G = |V(G)| - |E(G)|$.*

证明. Let G be a forest, and let its connected components be

$$G_1, \ldots, G_c,$$

where c is the number of connected components. Set

$$n_i := |V(G_i)|, \qquad m_i := |E(G_i)|.$$

Since G is a forest, each G_i is a tree. A basic fact about trees is that a tree on n_i vertices has exactly $n_i - 1$ edges, i.e.

$$m_i = n_i - 1 \qquad (1 \leq i \leq c).$$

Summing over all components gives

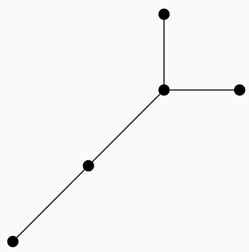
$$|E(G)| = \sum_{i=1}^{c} m_i = \sum_{i=1}^{c} (n_i - 1) = \sum_{i=1}^{c} n_i - c = |V(G)| - c.$$

Rearranging yields

$$c = |V(G)| - |E(G)|,$$

as claimed. □

Example 2.1.9. *Let T be a tree, $n := |V(T)| \geq 2$ with degree sequence $(d_1, \dots, d_n)$, ordered by non-increasing. For example:*

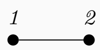


The degree sequence is $(3, 2, 1, 1, 1)$.

What is the number of trees on n vertices? $t(n)$. Cayley's formula solved this problem.

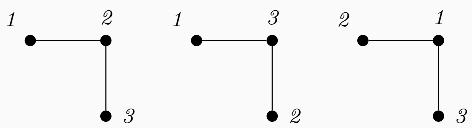
Example 2.1.10. *There are the basic examples:*

$n = 2$



$t(2) = 1$

$n = 3$



$t(3) = 3$

Theorem 2.1.11. $t(n) = n^{n-2}$.

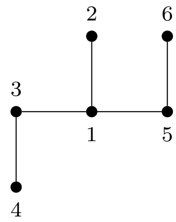
证明. **Proof 1:** Goal: find a bijection between trees on sets $[n]$ and $[n]^{n-2}$.

Let T be a tree

1. Pick $u_1 \in u = \min\{v \in V : \deg(v) = 1\}$ and $a_1 :=$ the unique neighbor of u_1.

2. Remove u_1, write down the lable of a_1, and repeat.

For example:



We pick $u_1 = 2$, $a_1 = 1$, and remove u_1. Then we pick $u_2 = 4$, $a_2 = 3$, and remove u_2. Then we pick $u_3 = 3$, $a_3 = 1$, and remove u_3. Then we pick $u_4 = 1$, $a_4 = 5$, and remove u_4. Now we get a sequence $a = (1, 3, 1, 5)$ in $[6]^4$.

After all, we get a sequence $a = (a_1, \ldots, a_{n-2})$ in $[n]^{n-2}$. Next we want to show: For eah sequence a in $[n]^{n-2}$, there exists a unique tree T.

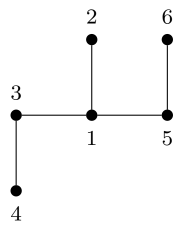
Let $\mathrm{d_i} := \sharp$ degree of vertex i and $f_i := \sharp$ occurances of i in a.

Note that $f_i \le d_i - 1$ for all i. Hence $n - 2 = \sum f_i \le \sum(d_i - 1) = (2n-2) - n = n - 2$. Hence $f_i = d_i - 1$ for all i.

Now we construct T:

1. For $i = 1, \ldots, n-2$: choose the smallest $b_i \in [n]$ not in $(a_i, a_{i+1}, \ldots, a_{n-2})$ and not equal to any b_j $(j < i)$; add edge (a_i, b_i).
2. So far we have $n - 2$ edges. The two vertices that never appear as $b_1, \ldots, b_{n-2}$ are exactly the last "remaining" pair; add the edge between them to get the $(n-1)$-th edge. Thus we obtain a tree T.

For example: given $a = (1, 3, 1, 5)$, we can construct the tree step by step :



Hence, we have a bijection between trees on sets $[n]$ and $[n]^{n-2}$. □

证明. **Proof 2:** Let $d_1, \ldots, d_n$ be positive integers with $\sum_{i=1}^{n} d_i = 2(n-1)$, and let

$$t(d_1, \ldots, d_n) := \#\{\text{labeled trees on } [n] = \{1, \ldots, n\} \text{ with } \deg(i) = d_i\}.$$

We first prove the degree-sequence formula

$$t(d_1, \ldots, d_n) = \binom{n-2}{d_1 - 1, \ldots, d_n - 1} = \frac{(n-2)!}{\prod_{i=1}^{n}(d_i - 1)!}. \qquad (*)$$

Step 1: a leaf-removal recurrence. Assume $d_n = 1$ (we may relabel so that one leaf is vertex n). In any such tree, the leaf n is adjacent to a unique vertex $i \in \{1, \ldots, n-1\}$. Removing n and its incident edge yields a labeled tree on $[n-1]$ whose degree sequence is

$$(d_1, \ldots, d_i - 1, \ldots, d_{n-1}).$$

Conversely, given a labeled tree on $[n-1]$ with degrees $(d_1,\dots,d_i-1,\dots,d_{n-1})$, attaching the new leaf n to i produces a labeled tree on $[n]$ with degrees $(d_1,\dots,d_{n-1},1)$. Hence

$$t(d_1,\dots,d_{n-1},1)=\sum_{i=1}^{n-1} t(d_1,\dots,d_i-1,\dots,d_{n-1}), \tag{R}$$

where terms with $d_i=1$ are understood to contribute 0.

Step 2: verification for the multinomial expression. Define

$$F(d_1,\dots,d_n):=\frac{(n-2)!}{\prod_{j=1}^{n}(d_j-1)!}.$$

For $d_n=1$ we have

$$F(d_1,\dots,d_{n-1},1)=\frac{(n-2)!}{\prod_{j=1}^{n-1}(d_j-1)!}.$$

Moreover, for $1\le i\le n-1$,

$$F(d_1,\dots,d_i-1,\dots,d_{n-1})=\frac{(n-3)!}{(d_i-2)!\prod_{j\neq i,\,j\le n-1}(d_j-1)!}=\frac{(n-3)!(d_i-1)}{\prod_{j=1}^{n-1}(d_j-1)!}.$$

Summing over i gives

$$\sum_{i=1}^{n-1} F(d_1,\dots,d_i-1,\dots,d_{n-1})=\frac{(n-3)!}{\prod_{j=1}^{n-1}(d_j-1)!}\sum_{i=1}^{n-1}(d_i-1).$$

Since $\sum_{i=1}^{n-1} d_i=2(n-1)-1=2n-3$, we have

$$\sum_{i=1}^{n-1}(d_i-1)=(2n-3)-(n-1)=n-2,$$

and therefore the right-hand side equals

$$\frac{(n-3)!(n-2)}{\prod_{j=1}^{n-1}(d_j-1)!}=\frac{(n-2)!}{\prod_{j=1}^{n-1}(d_j-1)!}=F(d_1,\dots,d_{n-1},1).$$

Thus F satisfies the same recurrence as t.

Step 3: induction. For $n=2$ there is exactly one labeled tree, so $t(1,1)=1=F(1,1)$. By induction on n using the recurrence, we conclude $t(d_1,\dots,d_n)=F(d_1,\dots,d_n)$.

Step 4: summing over degree sequences. Let $t(n)$ be the total number of labeled trees on $[n]$. Summing over all degree sequences with $\sum d_i=2(n-1)$ and setting $r_i=d_i-1$ yields

$$t(n)=\sum_{\substack{r_1,\dots,r_n\ge 0\\ r_1+\cdots+r_n=n-2}}\binom{n-2}{r_1,\dots,r_n}.$$

By the multinomial theorem,

$$(x_1+\cdots+x_n)^{n-2}=\sum_{r_1+\cdots+r_n=n-2}\binom{n-2}{r_1,\dots,r_n}x_1^{r_1}\cdots x_n^{r_n}.$$

Evaluating at $x_1 = \cdots = x_n = 1$ gives

$$n^{n-2} = \sum_{r_1+\cdots+r_n=n-2} \binom{n-2}{r_1,\ldots,r_n} = t(n).$$

Hence $t(n) = n^{n-2}$, which is Cayley's formula. □

证明. **Proof 3:** Using Kirchhoff's Matrix–Tree Theorem.

Let $t(n)$ denote the number of spanning trees of the complete graph K_n. Write A for the adjacency matrix and D for the diagonal degree matrix. The (combinatorial) Laplacian is

$$L(K_n) = D - A.$$

In K_n every vertex has degree $n-1$, hence $D = (n-1)I$. Moreover $A = J - I$, where J is the all-ones matrix. Therefore

$$L(K_n) = (n-1)I - (J - I) = nI - J.$$

We determine the eigenvalues of $L(K_n)$. Let $\mathbf{1} = (1,\ldots,1)^{\mathsf{T}}$. Since $J\mathbf{1} = n\mathbf{1}$, we have

$$L(K_n)\mathbf{1} = (nI - J)\mathbf{1} = n\mathbf{1} - n\mathbf{1} = 0,$$

so 0 is an eigenvalue (with eigenvector $\mathbf{1}$). If $x \in \mathbb{R}^n$ satisfies $\mathbf{1}^{\mathsf{T}}x = \sum_{i=1}^n x_i = 0$, then each entry of Jx equals $\sum_{i=1}^n x_i = 0$, hence $Jx = 0$, and thus

$$L(K_n)x = (nI - J)x = nx.$$

Consequently, n is an eigenvalue on the $(n-1)$-dimensional subspace $\mathbf{1}^{\perp} = \{x : \sum_i x_i = 0\}$, so the spectrum of $L(K_n)$ is

$$\underbrace{n,\ldots,n}_{n-1\text{ times}}, \quad 0.$$

By Kirchhoff's Matrix–Tree Theorem (eigenvalue form), if the Laplacian eigenvalues are $0 = \lambda_n < \lambda_1 \le \cdots \le \lambda_{n-1}$, then the number of spanning trees is

$$\tau(G) = \frac{1}{n}\prod_{i=1}^{n-1} \lambda_i.$$

Applying this to $G = K_n$ gives

$$t(n) = \tau(K_n) = \frac{1}{n} \cdot n^{n-1} = n^{n-2}.$$

This is Cayley's formula. □

第三章 Lecture 3

3.1 Chrometic number

Definition 3.1.1. *Let $G = (V, E)$ be a graph and let C be a set of colors. A* proper coloring *of G is a map*

$$f : V \to C$$

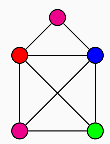
such that for every edge $\{x, y\} \in E$ (equivalently, for every pair of adjacent vertices $x \sim y$) we have

$$f(x) \neq f(y).$$

If G admits a proper coloring using at most k colors (that is, $|C| = k$), then G is called k-colorable.
The chromatic number *of G, denoted by $\chi(G)$, is the minimum number of colors needed for a proper coloring of G:*

$$\chi(G) = \min\{\, k : G \text{ is } k\text{-colorable}\}.$$

Example 3.1.2. *Consider following graph:*



We have $\chi(G) = 4$.

Corollary 3.1.3. *A graph G is bipartite if and only if $\chi(G) = 2$.*

证明. ($\Rightarrow$) Suppose G is bipartite. Then $V(G)$ can be written as a disjoint union

$$V(G) = A \cup B,$$

where every edge of G has one endpoint in A and the other in B. Define a coloring $f : V(G) \to \{1, 2\}$ by

$$f(v) = \begin{cases} 1, & v \in A, \\ 2, & v \in B. \end{cases}$$

If $\{u, v\} \in E(G)$, then u and v lie in different parts, hence $f(u) \neq f(v)$. Thus this is a proper coloring using two colors, so $\chi(G) \le 2$. Since G contains an edge, at least two colors are required, hence $\chi(G) = 2$.

($\Leftarrow$) Suppose $\chi(G) = 2$. Then there exists a proper coloring

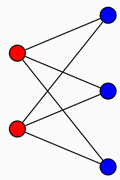
$$f : V(G) \to \{1, 2\}.$$

Let

$$A = f^{-1}(1), \qquad B = f^{-1}(2).$$

Then $V(G) = A \cup B$ and $A \cap B = \varnothing$. Since the coloring is proper, every edge $\{u, v\} \in E(G)$ satisfies $f(u) \ne f(v)$, so one endpoint lies in A and the other in B. Hence there are no edges inside A or inside B, and G is bipartite. □

Example 3.1.4. *Complete bipartite graph $K_{2,3}$:*



Theorem 3.1.5. *If G is planar then $\chi(G) \le 4$.*

Corollary 3.1.6. $\chi(K_n) = n$.

证明. In the complete graph K_n, every pair of distinct vertices is adjacent. Hence in any proper coloring, adjacent vertices must receive different colors, so all n vertices must have distinct colors. Thus $\chi(K_n) \ge n$.

On the other hand, assigning a different color to each vertex gives a proper coloring with n colors. Hence $\chi(K_n) \le n$.

Therefore $\chi(K_n) = n$. □

Corollary 3.1.7. *For a cycle C_n,*

$$\chi(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

证明. Let $V(C_n) = \{v_1, \dots, v_n\}$ where $v_i \sim v_{i+1}$ for $1 \le i < n$ and $v_n \sim v_1$.

If n is even, color the vertices alternately by two colors along the cycle:

$$f(v_i) = \begin{cases} 1, & i \text{ odd,} \\ 2, & i \text{ even.} \end{cases}$$

Since n is even, v_n and v_1 receive different colors, so this is a proper coloring. Hence $\chi(C_n) \le 2$. Since C_n has an edge, $\chi(C_n) \ge 2$, thus $\chi(C_n) = 2$.

If n is odd, alternating two colors along the cycle forces v_n and v_1 to have the same color, contradicting that they are adjacent. Hence $\chi(C_n) \geq 3$. Coloring the vertices alternately with two colors and giving v_n a third color yields a proper coloring with three colors, so $\chi(C_n) \leq 3$. Therefore $\chi(C_n) = 3$. □

Example 3.1.8. *consider the following graphs:*

Theorem 3.1.9 (Brooks' Theorem)**.** *Let $d \geq 3$ and let G be a graph with maximum degree d. If K_{d+1} is not a subgraph of G, then*

$$\chi(G) \leq d.$$

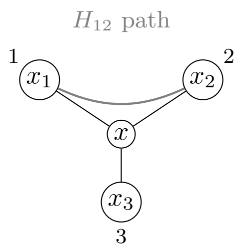
证明. Suppose, for a contradiction, that the theorem is false. Let G be a counterexample with the minimum number of vertices. Then $\chi(G) = d + 1$.

Let $x \in V(G)$ and let $N(x) = \{x_1, \ldots, x_\ell\}$, where $\ell \leq d$. Let $H = G - x$. By the minimality of G, we have $\chi(H) \leq d$. Take a proper d-coloring of H.

If fewer than d colors appear in $N(x)$, then we can color x with a color not used by its neighbors, obtaining a d-coloring of G, a contradiction. Hence $\ell = d$ and the vertices $x_1, \ldots, x_d$ receive pairwise distinct colors $1, \ldots, d$.

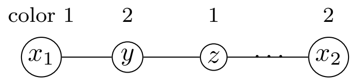
For $1 \leq i < j \leq d$, let H_{ij} be the subgraph of H induced by the vertices colored i or j. If x_i and x_j lie in different components of H_{ij}, then exchanging the colors i and j in the component containing x_i changes the color of x_i to j but leaves x_j unchanged. Thus color i disappears from $N(x)$, and we may color x with i, again obtaining a d-coloring of G, a contradiction.

Therefore, x_i and x_j lie in the same component of H_{ij} for all $i \neq j$. In particular, there exists an i–j path between them using only colors i and j.



If some vertex y in such a component had three neighbors in it, it would have at most $d - 3$ neighbors of other colors in H. Consequently, y is adjacent to at most $d - 2$ distinct colors in total. We can recolor y with a missing color $k \notin \{i, j\}$, which removes y from H_{ij}. Since x_i must have exactly one neighbor of color j in H_{ij} (otherwise its total degree in G would exceed d), the component of H_{ij} containing x_i and x_j must be a simple path (a Kempe chain); otherwise, recoloring a vertex of degree ≥ 3 would disconnect x_i and x_j, yielding the previous contradiction.

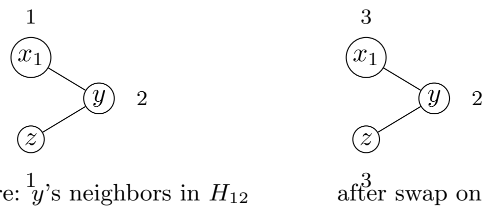
Since K_{d+1} is not a subgraph of G, the vertices in $N(x)$ cannot form a clique. Assume without loss of generality that x_1 and x_2 are not adjacent. The path H_{12} connecting x_1 and x_2 must contain at least one intermediate vertex. Let y be the neighbor of x_1 on H_{12}. The color of y is 2. Let z be the other neighbor of y on H_{12}, which has color 1.



path H_{12}

Now, consider the Kempe chain H_{13} connecting x_1 and x_3. We interchange the colors 1 and 3 on H_{13}. After this swap, x_1 receives color 3. The vertex y retains color 2, as it is not in H_{13}. To avoid a contradiction, x_1 and x_2 must still be connected by a new simple path H'_{23}. This path must start with the edge x_1y, meaning y must now have a second neighbor of color 3 to continue the path.

In the original coloring, y has exactly two neighbors of color 1 (x_1 and z) and exactly one neighbor of each remaining color $3, \ldots, d$. For y to gain a second neighbor of color 3 after the swap on H_{13}, its neighbor z must have belonged to H_{13} and changed its color from 1 to 3.



before: y's neighbors in H_{12}　　after swap on H_{13}

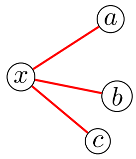
However, if $z \in H_{13}$, then z must have at least two neighbors of color 3 (to be in H_{13}), two neighbors of color 2 (to be in H_{12}), and at least one neighbor of each of the remaining $d - 3$ colors (to prevent it from being easily recolored). This requires $\deg(z) \geq 2 + 2 + (d - 3) = d + 1$, which contradicts $\max_{v \in V(G)} \deg(v) = d$.

This final contradiction demonstrates that no such minimal counterexample G can exist, completing the proof. $\square$

3.2 Ramsey theorem

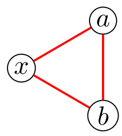
A classical motivation for Ramsey theory comes from the following social question posed in the 1950s: in any group of six people, there must exist either three people who all know each other or three people who are pairwise strangers.

This can be modeled using graph theory. Consider the complete graph K_6 whose vertices represent the six people. Color an edge red if the two corresponding people know each other and blue otherwise. Pick any vertex x. Among the five edges incident with x, at least three must have the same color (since $5 = 3 + 2$). Suppose, without loss of generality, that x has at least three red neighbors a, b, c.

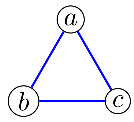


three red neighbors

If any pair among $\{a, b, c\}$ is joined by a red edge, say ab, then $\{x, a, b\}$ forms a red triangle. Otherwise the edges ab, bc, ca are all blue, and $\{a, b, c\}$ forms a blue triangle. Hence every red–blue coloring of the edges of K_6 contains a monochromatic triangle.



red triangle $\{x, a, b\}$



blue triangle $\{a, b, c\}$

This result is written as

$$R(3, 3) = 6,$$

where $R(s, t)$ denotes the smallest integer n such that every red–blue coloring of the edges of K_n contains either a red K_s or a blue K_t. The study of such unavoidable patterns in sufficiently large structures forms the basis of *Ramsey theory*.

$R(s, t)$ is the smallest integer n such that all edge colorings with 2 colors $\{c_1, c_2\}$ of K_n contain a monochromatic K_s in color c_1 or a monochromatic K_t in color c_2= smallest integer such that all graphs G on n vertices satisfies

(1) K_s is a subgraph of G or

(2) K_t is subgraph of $\bar{G}$

Corollary 3.2.1. *$R(2, t) = R(t, 2) = t$ for all $t \geq 2$.*

证明. We prove $R(2, t) = t$; the equality $R(t, 2) = t$ follows by symmetry.

Upper bound. In any red–blue coloring of $E(K_t)$, either there is a red edge, which is a red K_2, or else there are no red edges, in which case all edges are blue and the whole vertex set forms a blue K_t. Hence $R(2, t) \leq t$.

Lower bound. Consider K_{t-1} with all edges colored blue. Then there is no red K_2, and also no blue K_t (since there are only $t - 1$ vertices). Thus $R(2, t) \geq t$.

Therefore $R(2, t) = t$, and hence $R(t, 2) = t$. □

Proposition 3.2.2. $R(s, t) \leq R(s - 1, t) + R(s, t - 1)$

证明. Let

$$n := R(s - 1, t) + R(s, t - 1),$$

and consider any red–blue coloring of the edges of K_n. Fix a red vertex x. Let A be the set of red neighbors of x and B the set of blue neighbors of x. Then $|A| + |B| = n - 1$.

If $|A| \geq R(s-1,t)$, then the induced subgraph on A contains either a red K_{s-1} or a blue K_t. In the first case, together with x this forms a red K_s; in the second case we already have a blue K_t.

If $|A| < R(s-1,t)$, then

$$|B| = n - 1 - |A| \geq n - 1 - (R(s-1,t) - 1) = R(s,t-1).$$

as $n = R(s-1,t) + R(s,t-1)$ Hence the induced subgraph on B contains either a red K_s or a blue K_{t-1}. In the first case we already have a red K_s; in the second case, together with x this forms a blue K_t.

Thus every coloring of K_n contains a red K_s or a blue K_t, so $R(s,t) \leq n$. □

Corollary 3.2.3. $R(r,s) \leq \binom{r+s-2}{r-1}$.

证明. By induction on $r+s$. The base case $r+s=3$ is trivial. For the inductive step, assume the inequality holds for all $r+s' < r+s$. Then

$$R(r,s) \leq R(r-1,s) + R(r,s-1) \leq \binom{r+s-3}{r-2} + \binom{r+s-3}{s-2}.$$

By the inductive hypothesis,

$$\binom{r+s-3}{r-2} \leq \binom{r+s-2}{r-1} \quad \text{and} \quad \binom{r+s-3}{s-2} \leq \binom{r+s-2}{s-1}.$$

Hence $R(r,s) \leq \binom{r+s-2}{r-1}$. □

We first establish a recursive upper bound for the general Ramsey number $R(r,s)$, from which the diagonal bound follows.

Theorem 3.2.4 (Erdős-Szekeres Upper Bound)**.** *For any integers $s \geq 2$, the diagonal Ramsey number satisfies*

$$R(s,s) \leq \binom{2s-2}{s-1}.$$

Consequently, as $s \to \infty$, we have the asymptotic bound

$$R(s,s) \leq (1+o(1))\frac{4^{s-1}}{\sqrt{\pi s}}.$$

证明. The first part follows from the proof of the general bound $R(r,s) \leq \binom{r+s-2}{r-1}$.

Asymptotic Analysis: To derive the asymptotic behavior, we apply Stirling's approximation, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$:

$$\binom{2s-2}{s-1} = \frac{(2s-2)!}{((s-1)!)^2} \sim \frac{\sqrt{2\pi(2s-2)}\left(\frac{2s-2}{e}\right)^{2s-2}}{2\pi(s-1)\left(\frac{s-1}{e}\right)^{2s-2}} = \frac{4^{s-1}}{\sqrt{\pi(s-1)}}.$$

As $s \to \infty$, $\frac{1}{\sqrt{s-1}} = \frac{1}{\sqrt{s}}(1+o(1))$, yielding the desired asymptotic form:

$$R(s,s) \le (1+o(1))\frac{4^{s-1}}{\sqrt{\pi s}}.$$

□

This following theorem introduces the probabilistic method to combinatorics, demonstrating the existence of specific graphs by showing that the probability of their existence is strictly positive.

Theorem 3.2.5 (Erdős Exponential Lower Bound). *For any integer $n \ge s \ge 2$, if*

$$\binom{n}{s} 2^{1-\binom{s}{2}} < 1,$$

then $R(s,s) > n$. Consequently, we have the asymptotic lower bound:

$$R(s,s) \ge (1+o(1))\frac{s}{e\sqrt{2}} 2^{s/2}.$$

证明. Consider a complete graph K_n. We assign colors (red or blue) to each edge of K_n independently and uniformly at random with probability $\frac{1}{2}$.

Let S be a fixed subset of $V(K_n)$ of size s. Let A_S denote the event that the complete subgraph induced by S (i.e., K_s) is monochromatic (either all red or all blue). The probability of A_S occurring is:

$$\mathbb{P}(A_S) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{s}{2}} = 2^{1-\binom{s}{2}}.$$

Let X be the random variable counting the total number of monochromatic K_s subgraphs in K_n. We can express X as the sum of indicator variables: $X = \sum_{|S|=s} \mathbf{1}_{A_S}$. By the linearity of expectation, the expected number of monochromatic s-cliques is:

$$\mathbb{E}[X] = \sum_{|S|=s} \mathbb{P}(A_S) = \binom{n}{s} 2^{1-\binom{s}{2}}.$$

If $\mathbb{E}[X] < 1$, then by the properties of random variables, there must exist at least one point in the sample space (i.e., at least one specific 2-coloring of K_n) for which $X = 0$. Such a coloring contains strictly zero monochromatic K_s subgraphs. By definition, this implies that the Ramsey number $R(s,s)$ must be strictly greater than n.

Asymptotic Analysis: We seek the largest n such that the strict inequality holds. Using the standard upper bound for binomial coefficients $\binom{n}{s} \le \frac{n^s}{s!}$, it is sufficient to satisfy:

$$\frac{n^s}{s!} 2^{1-s(s-1)/2} < 1 \implies n^s < s! \cdot 2^{s(s-1)/2-1}.$$

Taking the s-th root of both sides gives:

$$n < (s!)^{1/s} \cdot 2^{(s-1)/2} \cdot 2^{-1/s}.$$

Applying Stirling's approximation $(s!)^{1/s} \sim \frac{s}{e}(\sqrt{2\pi s})^{1/s} \sim \frac{s}{e}$, and noting that $2^{(s-1)/2} = \frac{1}{\sqrt{2}}2^{s/2}$, we obtain:

$$n \approx \frac{s}{e} \cdot \frac{1}{\sqrt{2}}2^{s/2} = \frac{s}{e\sqrt{2}}2^{s/2}.$$

Therefore, setting n to be slightly less than this value guarantees the existence of a valid coloring, establishing the asymptotic lower bound:

$$R(s,s) \geq (1+o(1))\frac{s}{e\sqrt{2}}2^{s/2}.$$

□

Historical and Recent Progress

Result Type	**Year**	**Researchers**	**Key Breakthrough**
Classic Upper Bound	1935	Erdős, Szekeres	$R(s,s) \leq D \cdot 4^{s-1}$ (based on combinatorics).
Upper Bound Impr.	2009	David Conlon	Sub-exponential improvement: $R(s,s) \leq s^{-c\frac{\log s}{\log\log s}}4^s$.
Upper Bound Brkthr.	2023/24	Campos, Griffiths, Morris, et al.	First exponential improvement: $R(s,s) \leq (4-\epsilon)^s$ ($\epsilon \approx 2^{-7}$).
Classic Lower Bound	1947	Paul Erdős	$R(s,s) > C \cdot 2^{s/2}$ (dawn of the probabilistic method).
Lower Bound Brkthr.	2025	Jie Ma, W. Shen, S. Xie	Improvement for the off-diagonal Ramsey numbers $R(s,Cs)$.

For off-diagonal Ramsey numbers, we introduct the following results:

Result Type	**Year**	**Researchers**	**Key Contribution**
Classic Upper Bound	1935	Erdős, Szekeres	$r(s,t) = O(t^{s-1})$ for all fixed $s \geq 3$ and $t \to \infty$.
Upper Bound Impr.	1980	Ajtai, Komlós, Szemerédi	First improvement to the upper bound on $r(s,t)$ by analyzing a randomized greedy algorithm for producing large independent sets.
Lower Bound	2010	Bohman, Keevash	Lower bound by analyzing the random K_s-free graph process, improving on earlier results of Spencer.
Upper Bound Impr.	2001	Li, Rousseau, Zang	Extending ideas of Shearer, showed that $r(s,t) \leq (1 + o(1))\frac{t^{s-1}}{(\log t)^{s-2}}$ as $t \to \infty$.
Lower Bound Brk-thr.	1995	Kim	$r(3,t) = \Omega(t^2/\log t)$, matching previous upper bounds by Ajtai, Komlós and Szemerédi and Shearer, and improving earlier bounds of Spencer.
Asymptotic Result	2020	Fiz Pontiveros, Griffiths, Morris; Bohman, Keevash	$r(3,t)$ is determined asymptotically up to a factor of four.

第四章 Homework

4.1 Week 1

Exercise 4.1.1. *Let A be the adjacency matrix of a graph $G = (V, E)$.*

(a) Find a combinatorial interpretation for each entry of A^2 and A^3.

(b) Show that $\operatorname{tr}\left(A^2\right) - 2|E|$.

(c) For $m \geq 2$ an integer, what does $\operatorname{tr}\left(A^m\right)$ count? Prove your claim.

证明. (a) For $k \geq 1$ we have

$$(A^k)_{ij} = \sum_{t_1,\ldots,t_{k-1}} a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}.$$

Each product $a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}$ equals 1 precisely when

$$v_i \sim v_{t_1} \sim v_{t_2} \sim \cdots \sim v_{t_{k-1}} \sim v_j,$$

i.e. when $(v_i, v_{t_1}, \ldots, v_{t_{k-1}}, v_j)$ is a walk of length k from v_i to v_j; otherwise the product is 0. Hence $(A^k)_{ij}$ counts the number of walks of length k from v_i to v_j.

In particular:

$$(A^2)_{ij} = \sum_{t=1}^{n} a_{it} a_{tj}$$

counts the number of vertices v_t adjacent to both v_i and v_j, i.e. the number of length-2 walks $v_i \to v_t \to v_j$.

Similarly,

$$(A^3)_{ij} = \sum_{s,t=1}^{n} a_{is} a_{st} a_{tj}$$

counts the number of length-3 walks $v_i \to v_s \to v_t \to v_j$.

(b) Since $a_{ii} = 0$ for a simple graph, we have

$$(A^2)_{ii} = \sum_{t=1}^{n} a_{it} a_{ti} = \sum_{t=1}^{n} a_{it}^2 = \sum_{t=1}^{n} a_{it} = \deg(v_i).$$

Therefore

$$\operatorname{tr}(A^2) = \sum_{i=1}^{n} (A^2)_{ii} = \sum_{i=1}^{n} \deg(v_i).$$

By the Handshaking Lemma, $\sum_{i=1}^{n} \deg(v_i) = 2|E|$. Hence

$$\operatorname{tr}(A^2) = 2|E| \quad (\text{equivalently, } \operatorname{tr}(A^2) - 2|E| = 0).$$

(c) For any integer $m \geq 1$, by the argument in (a),

$$(A^m)_{ij} \text{ counts the number of walks of length } m \text{ from } v_i \text{ to } v_j.$$

In particular, $(A^m)_{ii}$ counts the number of *closed* walks of length m starting and ending at v_i. Summing over i yields

$$\operatorname{tr}(A^m) = \sum_{i=1}^{n} (A^m)_{ii},$$

so $\operatorname{tr}(A^m)$ counts the total number of closed walks of length m in G, where a closed walk is counted with its chosen starting vertex (i.e. each closed walk contributes 1 to exactly one diagonal entry, the entry corresponding to its start/end vertex).

□

Exercise 4.1.2. *The girth of a graph G is the length of the smallest polygon in the graph. Let G be a graph of girth 5 for which all vertices have degree at least d. Show that G has at least $d^2 + 1$ vertices. Can equality hold?*

证明. Fix a vertex $x \in V(G)$ and write $N(x) = \{y_1, \dots, y_t\}$ as the set of neighbors of x, where $t = \deg(x) \geq d$.

Step 1: no triangles. Since the girth is 5, G has no 3-cycles. Hence the neighbors of x are pairwise nonadjacent: $y_i \not\sim y_j$ for $i \neq j$.

Step 2: no 4-cycles. Again girth 5 implies there is no 4-cycle. Thus if $z \neq x$ were adjacent to both y_i and y_j for $i \neq j$, then $x - y_i - z - y_j - x$ would be a 4-cycle, a contradiction. Therefore the sets

$$N(y_i) \setminus \{x\} \qquad (i = 1, \dots, t)$$

are pairwise disjoint.

Step 3: counting. All vertices in

$$\{x\} \;\cup\; N(x) \;\cup\; \bigcup_{i=1}^{t} \big(N(y_i) \setminus \{x\}\big)$$

are distinct, so

$$|V(G)| \;\geq\; 1 + t + \sum_{i=1}^{t} \big|N(y_i) \setminus \{x\}\big| \;=\; 1 + t + \sum_{i=1}^{t} \big(\deg(y_i) - 1\big).$$

Since $\deg(y_i) \geq d$ for all i, we get $\deg(y_i) - 1 \geq d - 1$, hence

$$|V(G)| \;\geq\; 1 + t + t(d-1) \;=\; 1 + td \;\geq\; 1 + d^2.$$

This proves $|V(G)| \geq d^2+1$.
Step 4: the equality case. If $|V(G)| = d^2+1$, then all inequalities above are tight. In particular, $\deg(x) = t = d$ and $\deg(y_i) = d$ for all i. Repeating the argument with x replaced by an arbitrary vertex shows that G is d-regular. Tightness also forces that every vertex $z \notin \{x\} \cup N(x)$ is adjacent to *exactly one* neighbor of x (otherwise we would either miss vertices or create a 4-cycle). Equivalently, any two nonadjacent vertices have exactly one common neighbor, while adjacent vertices have none (otherwise a triangle would appear). Hence G is strongly regular with parameters

$$(v,k,\lambda,\mu) = (d^2+1,\ d,\ 0,\ 1).$$

Such graphs are precisely the Moore graphs of girth 5.
Step 5: why $d=4$ cannot have equality. Assume for contradiction that equality holds for $d=4$. Then G would be strongly regular with

$$(v,k,\lambda,\mu) = (17,4,0,1).$$

For a strongly regular graph, the two nontrivial adjacency eigenvalues are

$$r,s = \frac{(\lambda-\mu) \pm \sqrt{(\lambda-\mu)^2+4(k-\mu)}}{2},$$

and their multiplicities are

$$m_r = \frac{1}{2}\left((v-1) - \frac{2k+(v-1)(\lambda-\mu)}{\sqrt{\Delta}}\right), \qquad m_s = \frac{1}{2}\left((v-1) + \frac{2k+(v-1)(\lambda-\mu)}{\sqrt{\Delta}}\right),$$

where $\Delta = (\lambda-\mu)^2+4(k-\mu)$. Substituting $(v,k,\lambda,\mu) = (17,4,0,1)$ gives $\Delta = 13$ and $2k+(v-1)(\lambda-\mu) = 8+16(-1) = -8$, hence

$$m_r = \frac{1}{2}\left(16+\frac{8}{\sqrt{13}}\right), \qquad m_s = \frac{1}{2}\left(16-\frac{8}{\sqrt{13}}\right),$$

which are not integers. This contradicts that eigenvalue multiplicities must be integers. Therefore no such graph exists, and equality cannot hold when $d=4$. □

Remark 4.1.3. *Known examples achieving $|V(G)| = d^2+1$ occur for $d=2$ (the cycle C_5), $d=3$ (the Petersen graph), and $d=7$ (the Hoffman–Singleton graph). A hypothetical case $d=57$ is a famous open problem; for all other d equality is impossible.*

Exercise 4.1.4. *Let $Q = \{1,\ldots,q\}$ be the graph with vertex set Q^n, two vertices adjacent if they differ in exactly one coordinate. Show that Q is Hamiltonian.*

证明. We prove the statement by induction on n.

Base case $n=2$. The vertices are pairs $(a,b) \in Q^2$. Consider the cycle

$$(1,1),(2,1),\ldots,(q,1),(q,2),(q-1,2),\ldots,(1,2),(1,3),(2,3),\ldots,$$

continuing in this "serpentine" fashion through the rows $b = 1, 2, \ldots, q$, and finally ending at $(1, q)$ and returning to $(1, 1)$. Consecutive vertices differ in exactly one coordinate, and every vertex of Q^2 appears exactly once, so this is a Hamilton cycle in $H(2, q)$.

Inductive step. Assume $H(n-1, q)$ is Hamiltonian for some $n \geq 3$. Fix a Hamilton cycle in $H(n-1, q)$ and write it as

$$w_1, w_2, \ldots, w_m, w_1, \qquad \text{where } m = q^{n-1},$$

so that w_i, w_{i+1} differ in exactly one coordinate for each i (indices modulo m).

We now construct an explicit Hamilton cycle in $H(n, q)$. For each $i = 1, \ldots, m$, define a path P_i inside the fiber $\{(a, w_i) : a \in Q\} \subseteq Q \times Q^{n-1}$ by

$$P_i := \begin{cases} (1, w_i), (2, w_i), \ldots, (q-1, w_i), & \text{if } i \text{ is odd,} \\[2ex] (q-1, w_i), (q-2, w_i), \ldots, (1, w_i), & \text{if } i \text{ is even.} \end{cases}$$

Thus P_i visits exactly the vertices (a, w_i) with $a \in \{1, \ldots, q-1\}$, each exactly once, and consecutive vertices in P_i differ only in the first coordinate, hence are adjacent in $H(n, q)$.

Next, concatenate these paths in the order $P_1, P_2, \ldots, P_m$. The last vertex of P_i and the first vertex of P_{i+1} have the same first coordinate (both are $(q-1, *)$ or both are $(1, *)$ by construction), and their Q^{n-1}-parts are w_i and w_{i+1}, which differ in exactly one coordinate. Hence the connecting edge between P_i and P_{i+1} is also an edge of $H(n, q)$ (it changes only one of the last $n-1$ coordinates).

So far we have produced a Hamilton *path* through all vertices (a, w) with $a \in \{1, \ldots, q-1\}$ and $w \in Q^{n-1}$, i.e. through $(q-1)q^{n-1}$ vertices.

Finally, append the remaining q^{n-1} vertices with first coordinate q in the order

$$(q, w_m), (q, w_{m-1}), \ldots, (q, w_1),$$

and then close the cycle by returning to $(1, w_1)$. This works because:

- The last vertex of P_m is either $(1, w_m)$ or $(q-1, w_m)$, and in either case it is adjacent to (q, w_m) since only the first coordinate changes.
- For each j, the vertices (q, w_j) and (q, w_{j-1}) are adjacent because w_j and w_{j-1} differ in exactly one coordinate in $H(n-1, q)$.
- The vertex (q, w_1) is adjacent to $(1, w_1)$ since only the first coordinate changes.

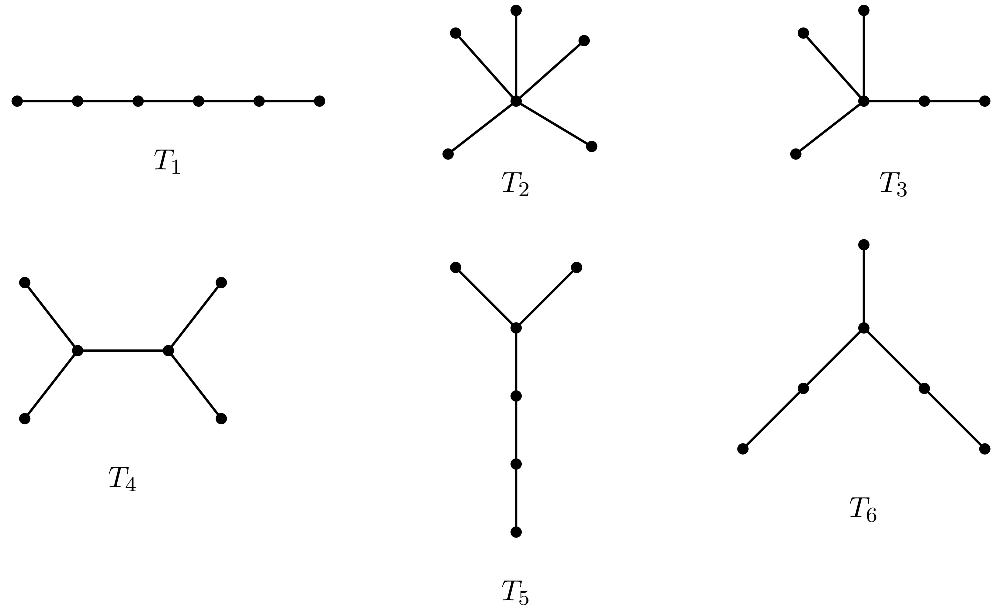
Thus we obtain a cycle in $H(n, q)$.

Moreover, every vertex of Q^n appears *exactly once* in this cycle: each (a, w_i) with $a \leq q-1$ appears exactly once in the concatenation of the P_i, and each (q, w_i) appears exactly once in the final appended segment. Hence the cycle is Hamiltonian.

This completes the induction and the proof. □

Exercise 4.1.5. *Find the six nonisomorphic trees on 6 vertices, and for each compute the number of distinct spanning trees in K_6 isomorphic to it.*

证明. One complete list of the six nonisomorphic trees on 6 vertices is shown below.



For the counting part, every spanning tree of K_6 is a labeled tree on the vertex set $\{1, 2, 3, 4, 5, 6\}$. Hence for each isomorphism type T_i,

$$N(T_i) = \frac{6!}{|\mathrm{Aut}(T_i)|},$$

where $\mathrm{Aut}(T_i)$ is the automorphism group of the unlabeled tree T_i.

Therefore:

$$\begin{aligned}
N(T_1) &= \frac{6!}{2} = 360, && \text{since } T_1 = P_6 \text{ has only the reversal symmetry;}\\
N(T_2) &= \frac{6!}{5!} = 6, && \text{since } T_2 = K_{1,5} \text{ allows an arbitrary permutation of its five leaves;}\\
N(T_3) &= \frac{6!}{3!} = 120, && \text{since the three leaves adjacent to the degree-4 vertex may be permuted;}\\
N(T_4) &= \frac{6!}{2\cdot 2\cdot 2} = 90, && \text{swap the two leaves on either side, interchange the two degree-3 vertices;}\\
N(T_5) &= \frac{6!}{2} = 360, && \text{since the two short branches at the degree-3 vertex may be interchanged;}\\
N(T_6) &= \frac{6!}{2} = 360, && \text{since the two branches of length 2 may be interchanged.}
\end{aligned}$$

So the numbers of spanning trees in K_6 isomorphic to $T_1, T_2, T_3, T_4, T_5, T_6$ are

$$360,\ 6,\ 120,\ 90,\ 360,\ 360.$$

As a check,

$$360 + 6 + 120 + 90 + 360 + 360 = 1296 = 6^{6-2},$$

which agrees with Cayley's formula for the total number of spanning trees of K_6. □

Exercise 4.1.6. *(a) Suppose a tree G has exactly one vertex of degree i for $2 \le i \le m$ and all other vertices have degree 1 . How many vertices does G have?*

(b) Suppose a tree G ha exactly one vertex of degree i for $2 \le i \le m$ and k other vertices have degree 1 . Prove that $k \ge \left\lfloor \frac{m+3}{2} \right\rfloor$ and give a construction for such a graph.

证明. (a) Let G have n vertices, where $m-1$ vertices have degrees $2, 3, \ldots, m$ respectively (each degree appears exactly once), and all other vertices have degree 1. Let k be the number of vertices of degree 1, then $n = (m-1) + k$.

By the Handshaking Lemma, the sum of degrees equals twice the number of edges. For a tree, the number of edges is $n-1$, so:

$$\sum_{v \in V} \deg(v) = 2(n-1)$$

Calculate the sum of degrees:

$$\sum_{v \in V} \deg(v) = 2 + 3 + \cdots + m + k \cdot 1 = \frac{m(m+1)}{2} - 1 + k$$

Therefore:

$$\frac{m(m+1)}{2} - 1 + k = 2(n-1) = 2(m+k-2)$$

Solving this equation:

$$\frac{m(m+1)}{2} - 1 + k = 2m + 2k - 4$$

$$\frac{m(m+1)}{2} - 1 + 4 = 2m + k$$

$$\begin{aligned} k &= \frac{m(m+1)}{2} - 2m + 3 \\ &= \frac{m^2 + m - 4m + 6}{2} \\ &= \frac{m^2 - 3m + 6}{2} \\ &= \frac{m(m-3)}{2} + 3 \end{aligned}$$

Thus the total number of vertices is:

$$n = (m-1) + k = m - 1 + \frac{m(m-3)}{2} + 3 = \frac{m^2 - m + 4}{2}$$

Answer: G has $\frac{m^2 - m + 4}{2}$ vertices.

(b) Let $G = (V, E)$ be a simple undirected graph with n vertices. According to the problem statement, G has exactly one vertex of degree i for each $2 \le i \le m$, and k vertices of degree 1. We want to show that $k \ge \lfloor \frac{m+3}{2} \rfloor$.

Order the vertices $v_1, v_2, \ldots, v_n$ in non-increasing order of their degrees. Thus, for $1 \le i \le m-1$, we have $\deg(v_i) = m - i + 1$, and the remaining k vertices have degree 1.

Let $c \geq 1$ be an integer. Define $H = \{v_1, v_2, \ldots, v_c\}$ as the set of the c vertices with the highest degrees. The sum of the degrees of the vertices in H is:

$$\begin{aligned} S_H = \sum_{j=1}^{c} \deg(v_j) &= m + (m-1) + \cdots + (m-c+1) \\ &= \frac{c(2m-c+1)}{2} \end{aligned}$$

The degrees of vertices in H are contributed by edges within H and edges between H and $V \setminus H$. Let $E(H)$ be the number of edges with both endpoints in H, and $E(H, \bar{H})$ be the number of edges with one endpoint in H and the other in $V \setminus H$. By double counting, we have:

$$S_H = 2|E(H)| + |E(H, \bar{H})|$$

Since G is a simple graph, the number of internal edges is at most $|E(H)| \leq \frac{c(c-1)}{2}$. Furthermore, any vertex $u \in V \setminus H$ can be adjacent to at most $\min(\deg(u), c)$ vertices in H. Therefore:

$$S_H \leq 2 \cdot \frac{c(c-1)}{2} + \sum_{u \in V \setminus H} \min(\deg(u), c)$$

We choose $c = \lfloor \frac{m+1}{2} \rfloor$. Under this choice, the maximum degree in $V \setminus H$ is $m - c \leq c$. Thus, for every $u \in V \setminus H$, we have $\min(\deg(u), c) = \deg(u)$. The sum of degrees of vertices in $V \setminus H$ splits into the remaining higher-degree vertices and the k leaves:

$$\sum_{u \in V \setminus H} \deg(u) = \sum_{j=c+1}^{m-1} (m - j + 1) + k \cdot 1 = \frac{(m-c-1)(m-c+2)}{2} + k$$

Substituting the expressions for S_H and the upper bounds into our inequality yields:

$$\frac{c(2m-c+1)}{2} \leq c(c-1) + \frac{(m-c-1)(m-c+2)}{2} + k$$

Multiplying by 2 and expanding both sides:

$$2cm - c^2 + c \leq 2c^2 - 2c + (m^2 - 2cm + c^2 + m - c - 2) + 2k$$

Isolating $2k$, we obtain a quadratic inequality in terms of c:

$$2k \geq -4c^2 + 4(m+1)c - (m^2 + m - 2)$$

By completing the square for the quadratic expression in c, we get:

$$2k \geq -4\left(c - \frac{m+1}{2}\right)^2 + m + 3$$

We now evaluate this lower bound based on the parity of m:

- **Case 1: m is odd.** We chose $c = \lfloor \frac{m+1}{2} \rfloor = \frac{m+1}{2}$. Substituting this into the inequality gives:

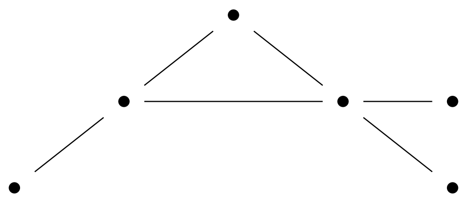
$$2k \geq 0 + m + 3 \implies k \geq \frac{m+3}{2} = \left\lfloor \frac{m+3}{2} \right\rfloor$$

- **Case 2: m is even.** We chose $c = \lfloor \frac{m+1}{2} \rfloor = \frac{m}{2}$. Substituting this yields $c - \frac{m+1}{2} = -\frac{1}{2}$, so:

$$2k \geq -4\left(-\frac{1}{2}\right)^2 + m + 3 = m + 2 \implies k \geq \frac{m+2}{2} = \left\lfloor \frac{m+3}{2} \right\rfloor$$

In both cases, we conclude that $k \geq \lfloor \frac{m+3}{2} \rfloor$. This completes the proof.

Besides, an example can be constructed as following ($m = 4$):



$\square$